

Name:R.K Ayushman Singh

Reg no:261100690014

LOS Assignment Theory-5

command line arguments, fileoperators, read and filters

Write a Bash script that accepts a filename as the first command-line argument.

Check whether an argument is provided.

Check whether the file exists using a file operator.

If the file exists, display the first 5 lines using head.

Otherwise, display an appropriate error message.

```
anonymous@anonymous-supreme in ~/code as 🐼 took 24s824ms
➤ ./showfile.sh demo.txt
First 5 lines of demo.txt:
hi
anonymous
supreme warrior
yo
hhee

anonymous@anonymous-supreme in ~/code as 🐼
➤ cat showfile.sh
File: showfile.sh
#!/bin/bash

if [ $# -eq 0 ]; then
    echo "Error: No filename provided."
    exit 1
fi

if [ -e "$1" ]; then
    echo "First 5 lines of $1:"
    head -n 5 "$1"
else
    echo "Error: File does not exist."
fi

anonymous@anonymous-supreme in ~/code as 🐼
➤
```

File Type and Permission Checker

Write a Bash script that accepts a path as a command-line argument and determines whether it is:

a regular file,

a directory,

readable,

writable, and

executable.

Display suitable messages using file operators such as -f, -d, -r, -w, and -x.

```
anonymous@anonymous-supreme in ~/code
[❯] chmod +x checkfile.sh

anonymous@anonymous-supreme in ~/code
[❯] ./checkfile.sh demo.txt
It is a regular file.
It is readable.
It is writable.
It is not executable.

anonymous@anonymous-supreme in ~/code
[❯] cat checkfile.sh
File: checkfile.sh
#!/bin/bash

if [ $# -eq 0 ]; then
    echo "Error: No path provided."
    exit 1
fi

if [ -f "$1" ]; then
    echo "It is a regular file."
fi

if [ -d "$1" ]; then
    echo "It is a directory."
fi

if [ -r "$1" ]; then
    echo "It is readable."
else
    echo "It is not readable."
fi

if [ -w "$1" ]; then
    echo "It is writable."
else
    echo "It is not writable."
fi

if [ -x "$1" ]; then
    echo "It is executable."
else
    echo "It is not executable."
fi

anonymous@anonymous-supreme in ~/code
[❯]
```

Write a Bash script that accepts a log file as a command-line argument and searches for the words:

failed, denied, unauthorized, and error.

```
anonymous@anonymous-supreme in ~/code
# > nano log.txt

anonymous@anonymous-supreme in ~/code took 7s452ms
# > nano searchlog.sh

anonymous@anonymous-supreme in ~/code
# > chmod +x searchlog.sh

anonymous@anonymous-supreme in ~/code
# > ./searchlog.sh log.txt
ERROR: Unable to connect to database
FAILED: Login attempt failed
WARNING: Access denied
UNAUTHORIZED: Unauthorized access detected
ERROR: Network connection failed

anonymous@anonymous-supreme in ~/code
# > cat searchlog.sh
File: searchlog.sh
#!/bin/bash

if [ $# -eq 0 ]; then
    echo "Error: No log file provided."
    exit 1
fi

if [ ! -f "$1" ]; then
    echo "Error: Log file does not exist."
    exit 1
fi

grep -i -E "failed|denied|unauthorized|error" "$1"

anonymous@anonymous-supreme in ~/code
# >
```

Write a Bash script that accepts a filename and checks whether:

the file exists,

it is a regular file,

it is empty or non-empty.

If it is non-empty, display the first 5 lines.

```
anonymous@anonymous-supreme in ~/code took 4s597ms
# ) bash

anonymous@anonymous-supreme in ~/code
# ) nano filecheck.sh

anonymous@anonymous-supreme in ~/code
# ) chmod +x filecheck.sh

anonymous@anonymous-supreme in ~/code
# ) ./filecheck.sh log.txt
File exists.
It is a regular file.
File is non-empty.
First 5 lines:
INFO: System started successfully
ERROR: Unable to connect to database
INFO: User logged in
FAILED: Login attempt failed
WARNING: Access denied
```

Write a Bash script named investigate.sh that accepts a log file as a command-line argument.

The script must first verify that the file exists and is readable. Then ask the investigator to enter a search term.

Display the first 5 matching records and report the total number of matching records.

```
anonymous@anonymous-supreme in ~/code took 33s268ms
# ) nano investigate.sh

anonymous@anonymous-supreme in ~/code took 3s112ms
# ) chmod +x investigate.sh

anonymous@anonymous-supreme in ~/code
# ) ./investigate.sh log.txt
Enter search term: error
First 5 matching records:
2026-09-19 09:02:45 ERROR Database connection failed
2026-09-19 09:06:33 ERROR Network connection failed
2026-09-19 09:10:27 ERROR File not found: config.txt
2026-09-19 09:13:52 ERROR Disk read error
2026-09-19 09:19:31 ERROR Service failed to start
Total matching records: 5

anonymous@anonymous-supreme in ~/code took 3s821ms
```